

Lecture 2 Introduction to Linux Notes (this is the title of the document - Heading 1 formatting please!)

1. What is an Operating System?

An operating system (OS) is the core software that manages a computer's hardware and runs applications. It acts as a bridge between you and the machine.

2. What is a kernel?

A kernel is the core computer program at the center of an operating system that acts as a bridge between software applications and physical hardware

3. Which other parts aside from the kernel identify an OS?

An operating system includes user-space components like the shell, system utilities, and application programming interfaces (APIs) that work alongside the kernel to define and run the system.

4. What is Linux?

Linux is a free and open-source operating system that manages computer hardware and software

5. What is a Linux distribution?

A Linux distribution (or distro) is a complete operating system built around the Linux distribution kernel, combined with software tools, system libraries, and a package manager.

6. List at least 4 Linux characteristics:

- **Open-Source:** The source code is freely available for anyone to view, modify, and distribute under the GNU General Public License.
- **Multi-User:** Multiple users can access system resources, memory, and applications at the same time without interfering with each other.
- **Multitasking:** The operating system can run multiple programs and processes concurrently by sharing CPU time efficiently.
- **High Stability and Reliability:** Linux systems rarely crash and can run continuously for years, making them ideal for enterprise servers and critical infrastructure.

7. What is Debian?

Debian is a free, open-source operating system built on the Linux kernel and developed by a worldwide community of volunteers.

8. List and define the different types of licensing agreements

Licensing agreements are broken down by their level of exclusivity and the type of intellectual property involved.

Types by Exclusivity

- Exclusive License: Gives one single licensee the total right to use the IP. Even the original owner (licensor) cannot use the IP within that defined scope or territory.
- Non-Exclusive License: Allows the licensor to grant the same IP rights to multiple licensees at the same time. This works well for widespread product or software distribution.
- Sole License: A middle-ground option where only one licensee gets the rights, but the licensor keeps the right to use the asset as well.

Types by Intellectual Property (IP)

- Patent Licensing: Allows a company to manufacture, use, or sell a patented invention, design, or manufacturing process created by another party.
- Trademark Licensing: Authorizes a third party to use a business brand name, logo, or slogan on products or services while the owner keeps brand ownership.
- Copyright Licensing: Grants permission to reproduce, distribute, perform, or adapt creative works—such as music, art, books, or films—while ownership stays with the creator.
- Software Licensing: Governs how users can access, install, or run software. Common models include subscription-based access or traditional perpetual (one-time purchase) rights.
- Trade Secret Licensing: Gives controlled, confidential access to valuable business processes, formulas, or algorithms that derive economic value from remaining secret.

9. What is Free Software? Define the 4 freedoms.

Free software is a matter of liberty, not price—think "free as in free speech, not as in free beer". According to the Free Software Foundation, a program is free software if its users have ultimate control over the software and its function. You might pay money to get a copy of free software or get it for zero cost, but you always keep the freedom to share and change it.

The 4 Essential Freedoms

- Freedom 1: The freedom to run the program as you wish, for any purpose.
- Freedom 2: The freedom to study how the program works and change it, which requires access to the source code.
- Freedom 3: The freedom to redistribute copies to help others.
- Freedom 4: The freedom to distribute modified versions to the community, which also requires access to the source code.

10. What is virtualization?

Virtualization is a technology that uses software to create a simulated, or virtual, version of a physical computer hardware system, server, storage device, or network.